

14/9/25

Assignment - 1

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Section - A

1) What are the features of Data Warehousing?

Ans Subject Oriented → Organized by subject like sales, customer, products.

• Integrated → Data is collected from different source and made consistent.

• Time - variant → Store historical data from analysis over time.

• Non-volatile → Data is stable, once stored, it is not frequently changed.

2) What is the difference between Data Mining and Data Warehouses.

Ans Data Warehousing → collect and store data from many source for analysis.

Data Mining → Find patterns, trends, useful information from stored data.

3) What are the benefits of data mining?

Ans → Helps in better decision-making.

→ Find hidden patterns and trends.

→ Improve marketing, sales, customer satisfaction.

→ Detect fraud and risks.

4) What is Data Mining?

Ans Data mining is the process of analysing

large sets of data to discover useful patterns, trends and relationships.

5) What type of information is collected in Data Mining?

Ans: Pattern and trends.

- Customer behavior.
- Prediction (future sales and risks)
- Classification and clustering of data.

Section - B

1) Discuss all steps of Data Warehouse process.

Ans: The process of building and using a data warehouse involve several steps:-

1. Data Collection → Data is collected from multiple sources such as database, files, ERP, CRM etc.
2. Data Cleaning → Errors, missing values and duplicates are removed so that data becomes accurate and reliable.
3. Data Transformation → Data is converted into a standard format, units are made consistent and values are normalized.
4. Data Loading → Transformed data is loaded

into data warehouse using (ETL) Extract, Transform, Load tools.

5) Data Storage - The warehouse store historical and current data in organized manner.

6) Data Access and Analysis - Users use OLAP tool, queries, report to analysis the data for decision making.

2) What kind of data can be mined?

Ans. • Relational Data - Data store in tables of Database (rows and column).

- Transactional Data - Data from transaction or activities. ex -> online shopping history
- Text Data - Data in Document, email, articles
- Multiple media Data - Image, audio, video-graphics eg. Face Recognition from photo, video.
- Time-series and Sequence Data - Data collected over time or in sequence
ex:- stock market, weather data
- Spatial Data - Data related to geography and location eg -> maps, satellite image.

7) Web Data → Data from the internet like clickstream, logs, social media.

8) Discuss Functionality of Data mining.
Ans Functionality of data mining →

1. Classification → Assign data into predefined categories.
eg :- classify emails into spam or not spam.

2) Clustering → group similar data without predefined labels.

3) Associating Rule Mining → Find relationship among items.

ex:- if a customer buys bread they often buy butter.

4) Prediction → Forecast future values based on current data.
eg - Prediction stock price.

5) Regression → Estimates numerical values.
eg → predicting house prices based on size and location.

6) Outlier → Find unusual data that does not fit normal pattern.
eg → Detecting fraudulent credit card transaction.

Section - C

- 1) Define KDD. Identify and describe phase in KDD processes.

Ans KDD (Knowledge Discovery in Database)

KDD is the overall process of discovering useful knowledge, hidden pattern and relationships from large amount of data. Data mining is one important step in the KDD process.

Phases

1. Data selection - Relevant data is chosen from multiple source.
2. Data cleaning and preprocessing - Noise, missing value, and errors are removed.
3. Data Transformation - Data is converted into proper format and reduced in size if necessary.
4. Data mining - Core step where algorithms are applied to extract patterns.
5. Pattern evaluation - Interesting and useful pattern are selected from the mined data.
6. Knowledge presentation - Result are presented to users using visualization and report.

- 2) Discuss Architecture of data Warehouse with diagram.

Ans A Data Warehouse architecture shows how

data flows from source into the warehouse and then to users.

Main Component

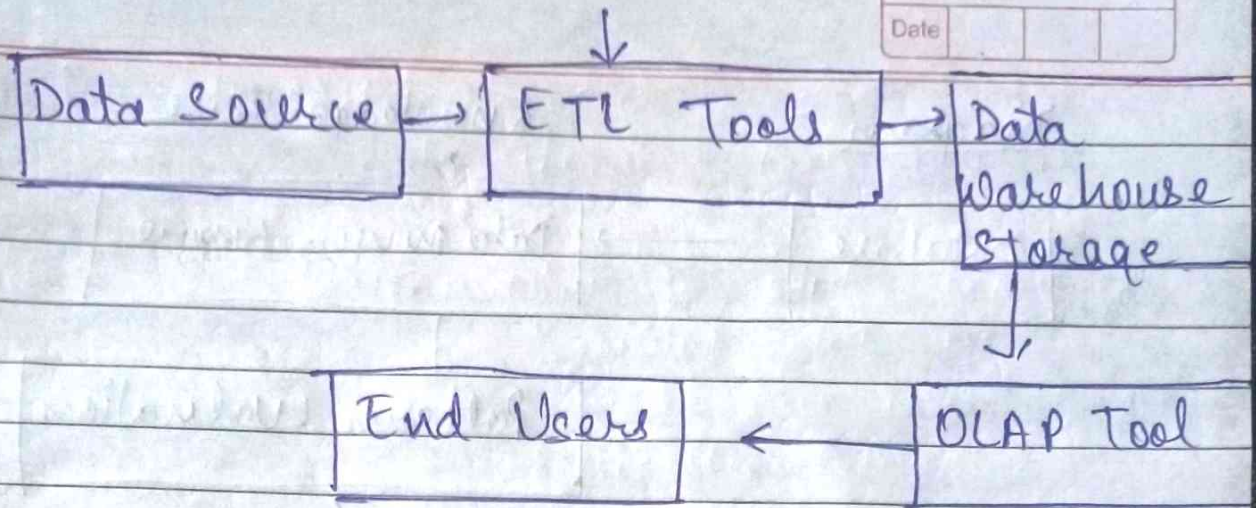
1. Data source → Different operational database, flat files, external data (ERP, CRM, Excel).
2. ETL Process (Extract Transform Load)
 - Extract → collect data from source.
 - Transform: cleans, format, and joins integrates data.
 - Load: Insert data into the warehouse.
3. Data warehouse storage → Central repository that stores integrated and historical data.
4. Meta data → 'Data about data' that describe the source, format, transformation and schema.
5. OLAP Engine / Tools → Online Analytical processing tool used to analyze, slice, dice, roll-up and drill-down data.
6. End Users / Client → Business analysts, managers, decision-makers use queries, dashboard and reports.

Diagram

Meta Data

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3. Discuss Architecture of Data mining system.

Ans The data mining system architecture shows how data is processed to extract useful patterns.

Main Component

1. Database / Data Warehouse → Store large amount of data
2. Database / Knowledge base → Stores useful rules, prior knowledge and constraints.
3. Data Mining Engine → The core component that performs tasks like classification, clustering, association and prediction.
4. Pattern Evaluation Module → Filters and select the interesting, useful and valid patterns from result.
5. User Interface → Provide Interaction b/w the system and users, allows queries, visualization and result interpretation

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Diagram

Knowledge Base

Database

Data Mining Engine

Pattern Evaluation

User Interface